

Trypanosomiasis.

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African (sleeping sickness) and American (Chagas' disease) trypanosomiasis, caused by protozoa of the family Trypanosomatidae, are diseases that are endemic in parts of Africa and Latin America, respectively. Physicians in developed countries may occasionally see cases because of extensive travel and immigration from endemic countries. Although neurological involvement is common in both, its incidence and clinical presentation differ considerably. African trypanosomiasis, caused by subspecies of *Trypanosoma brucei* (*T b rhodesiense*, *T b gambiense*), is transmitted by the tsetse fly and causes meningoencephalitis, in which somnolence is a prominent feature. Parasites may reach the brain parenchyma through the choroid plexus or the Virchow Robin spaces. American trypanosomiasis, caused by *Trypanosoma cruzi* is transmitted by reduviid bugs. While lesions in the central nervous system are not prominent, except in the reactivated forms which occur in immunodeficient patients, the peripheral nerve, mainly the autonomic nervous system, is frequently involved, leading to the cardiomegaly and the digestive megaviscera. Congenital infections may also occur. In this paper we give an account of the epidemiology, clinical presentation and pathological features of these two protozoal infections based on human and experimental studies of both the central and peripheral nervous system.

Human African trypanosomiasis.

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Human African trypanosomiasis (sleeping sickness) is caused by the unicellular parasite *Trypanosoma brucei* and transmitted by tsetse flies. It occurs exclusively in sub-Saharan Africa, usually in rural areas affected by civil conflicts and neglected health systems. Reported cases are fewer than 10,000/year, which classifies it as one of the most neglected tropical diseases. Because sleeping sickness is fatal if not treated, it has to be included in the differential diagnosis of every febrile traveler returning from a game park in East Africa. Elimination of the disease is considered feasible provided better tools for diagnosis and treatment can be made available.

[Towards elimination of human African trypanosomiasis].

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Human African Trypanosomiasis (HAT) is caused by *Trypanosoma brucei* which is transmitted by the tsetse fly insect vector (*Glossina* spp). It is one of the 20 Neglected Tropical Diseases (NTD) listed by the WHO. These diseases affect the poorest and most vulnerable communities, for which the WHO has established a dedicated 2021-2030 roadmap. At the time of Alphonse Laveran, HAT devastated the African continent. In the 1960s, the disease was nearly under control, but it strongly re-emerged in the 1990s. A coordinated effort of all stakeholders, with national control programs as the main actors, a strong contribution of research and important donations by the private sector, allowed to decrease the HAT burden significantly. Since 2018, less than 1000 cases are detected annually. We here review new diagnostics, treatments and vector control tools that have been implemented jointly and successfully in several endemic countries. The next key challenge will be to sustain the gains. Newly emerging research questions include long-term carriage of trypanosomes and adaptation of tools to low prevalence

contexts. Challenges out of the research area comprise the continued need of funding, maintenance of dedicated human resources, and the key question of access. Sustainable elimination as "interruption of transmission", which is the 2030 NTD roadmap target, can be reached, if these challenges are solved. We stress the importance of continuing to combine the efforts in the fight against the disease, because sustainable elimination of HAT is the best long-term prevention strategy against re-emergence. As such, HAT elimination can serve as an example for other infectious diseases.

The epidemiology and control of human African trypanosomiasis.

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Human African trypanosomiasis is caused by *Trypanosoma brucei gambiense* in West and Central Africa, and by *Trypanosoma brucei rhodesiense* in East and southern Africa. In recent years there has been a dramatic resurgence of Gambian trypanosomiasis in Central Africa, especially in the Democratic Republic of Congo, Angola and Sudan. The disease is quiescent in most of West Africa, as is Rhodesian trypanosomiasis the other side of the continent. The epidemiology of Gambian trypanosomiasis is reviewed in detail. The long duration of infection in human hosts with cycles of intermittent parasitaemia, the vectors' feeding habits and the intensity of human-fly contact are the major determinants of the dynamics of transmission of this parasite. The development of immunity may lead to a reduction in the fraction of the population that is susceptible to infection and the burning out of epidemics after 20 to 30 years. So far, the acquired immune deficiency syndrome pandemic has had no impact on the epidemiology of Gambian trypanosomiasis. A brief review of the epidemiology of Rhodesian trypanosomiasis highlights the differences from Gambian trypanosomiasis that, to some extent, explain its lower propensity to cause epidemics: it is a more aggressive disease that rapidly kills its human host, and its transmission involves mostly domestic and game animals, humans being in most circumstances an accidental host. The various methods and strategies for the surveillance and control of both diseases are reviewed.

The epidemiology of trypanosomiasis in Rumphi district, Malawi: a ten year retrospective study.

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Human African Trypanosomiasis (HAT) is caused by two species of the tsetse fly vectored protozoan hemoflagellates belonging to *Trypanosma brucei*, namely *T.b gambiense* which predominates in Western Africa and follows a chronic disease course and *T.b rhodensiense* which is more prevalent in Southern and Eastern Africa, Malawi included, and follows a more acute and aggressive disease course. Previous studies in the Democratic Republic of Congo, Angola, Uganda and Sudan have demonstrated that the prevalence rates of *T.b rhodensiense* infection have reached epidemic proportions. To describe the epidemiology of Trypanosomiasis in Rumphi District over the past ten years. A total of 163 records from January 2000 to December 2006 were retrospectively studied. There were more males than females (121 vs. 40) with the 20 - 29 years age bracket having the highest number of cases (26.3%, n = 160). Stage 2 HAT was the commonest stage at presentation (58.2%, n = 158) with the patients in the same being 3.5 times more likely to die than those with stage 1 HAT. Case fatality rates for late and early stage disease were 21.5% (n = 92) and 7.2% (n = 66) respectively with 84.6% having been cured (n = 162).

Convulsions were associated with fatal disease outcome and the majority of cases (97.2%, $n = 103$) lived within 5 kilometres of the Vwaza game reserve boundary. More men have been infected than women, with a high involvement in the 20 - 29 age brackets. A dramatic increase with active case finding indicates a high under-detection of the disease with late stage HAT being predominant at presentation. Though it has been found that cases with late stage disease have an increased likelihood of dying compared to those in early stage HAT, the high proportion of successful treatment indicates that the disease still carries a high degree of favourable outcome with treatment. It has also been demonstrated in this study that more than 95% of trypanosomiasis cases live within 5 km of game reserve boundary. Disease interventions should be implemented in areas within 5 km of marshland game reserve boundary as priority areas.

Human African trypanosomiasis.

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Human African trypanosomiasis or sleeping sickness is a neglected tropical disease that affects populations in sub-Saharan Africa. The disease is caused by infection with the gambiense and rhodesiense subspecies of the extracellular parasite *Trypanosoma brucei*, and is transmitted to humans by bites of infected tsetse flies. The disease evolves in two stages, the hemolymphatic and meningoencephalitic stages, the latter being defined by central nervous system infection after trypanosomal traversal of the blood-brain barrier. African trypanosomiasis, which leads to severe neuroinflammation, is fatal without treatment, but the available drugs are toxic and complicated to administer. The choice of medication is determined by the infecting parasite subspecies and disease stage. Clinical features include a constellation of nonspecific symptoms and signs with evolving neurological and psychiatric alterations and characteristic sleep-wake disturbances. Because of the clinical profile variability and insidiously progressive central nervous system involvement, disease staging is currently based on cerebrospinal fluid examination, which is usually performed after the finding of trypanosomes in blood or other body fluids. No vaccine being available, control of human African trypanosomiasis relies on diagnosis and treatment of infected patients, assisted by vector control. Better diagnostic tools and safer, easy to use drugs are needed to facilitate elimination of the disease.

Target product profile: *Trypanosoma brucei gambiense* test to verify elimination.

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Human African trypanosomiasis is a life-threatening parasitic infection transmitted by the tsetse fly in sub-Saharan Africa. The most common form is caused by *Trypanosoma brucei gambiense*, with humans as the main reservoir. Diagnosis in the field requires microscopic examination performed by specifically trained personnel. After over two decades of sustained efforts, the incidence of the disease is strongly declining, and some historically endemic countries are no longer detecting cases. The World Health Organization (WHO) has targeted the elimination of transmission of gambiense human African trypanosomiasis by 2030, defined as zero autochthonous cases for at least five consecutive years. Endemic countries reaching this goal must maintain dedicated surveillance to detect re-emergence or re-introduction. With this new agenda, new tools are needed for verification of the absence of transmission. WHO has therefore developed a target product profile calling for development of a method for population-level cross-cutting surveillance of *T. b. gambiense* transmission. The method needs to be

performed in national or sub-national reference laboratories, and to test in parallel numerous samples shipped from remote rural areas. Among other characteristics the product profile specifies: (i) a simple specimen collection procedure; (ii) no cold-chain requirement to transfer specimens to reference laboratories; (iii) high sensitivity and specificity; (iv) high-throughput, substantially automatized; (v) low cost per specimen, when analysed in large batches; and (vi) applicable also in animals.

Overview of the Diagnostic Methods Used in the Field for Human African Trypanosomiasis: What Could Change in the Next Years?

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Sleeping sickness is a parasitic infection caused by two species of trypanosomes (*Trypanosoma brucei gambiense* and *rhodesiense*), transmitted by the tsetse fly. The disease eventually affects the central nervous system, resulting in severe neurological symptoms. Without treatment, death is inevitable. During the first stage of the disease, infected patients are mildly symptomatic and early detection of infection allows safer treatment (administered on an outpatient basis) which can avoid death; routine screening of the exposed population is necessary, especially in areas of high endemicity. The current therapeutic treatment of this disease, especially in stage 2, can cause complications and requires a clinical surveillance for several days. A good stage diagnosis of the disease is the cornerstone for delivering the adequate treatment. The task faced by the medical personnel is further complicated by the lack of support from local health infrastructure, which is at best weak, but often nonexistent. Therefore it is crucial to look for new more efficient technics for the diagnosis of stage which are also best suited to use in the field, in areas not possessing high-level health facilities. This review, after an overview of the disease, summarizes the current diagnosis procedures and presents the advances in the field.

Epidemiology of human African trypanosomiasis.

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Human African trypanosomiasis (HAT), or sleeping sickness, is caused by *Trypanosoma brucei gambiense*, which is a chronic form of the disease present in western and central Africa, and by *Trypanosoma brucei rhodesiense*, which is an acute disease located in eastern and southern Africa. The *rhodesiense* form is a zoonosis, with the occasional infection of humans, but in the *gambiense* form, the human being is regarded as the main reservoir that plays a key role in the transmission cycle of the disease. The *gambiense* form currently assumes that 98% of the cases are declared; the Democratic Republic of the Congo is the most affected country, with more than 75% of the *gambiense* cases declared. The epidemiology of the disease is mediated by the interaction of the parasite (trypanosome) with the vectors (tsetse flies), as well as with the human and animal hosts within a particular environment. Related to these interactions, the disease is confined in spatially limited areas called "foci", which are located in Sub-Saharan Africa, mainly in remote rural areas. The risk of contracting HAT is, therefore, determined by the possibility of contact of a human being with an infected tsetse fly. Epidemics of HAT were described at the beginning of the 20th century; intensive activities have been set up to confront the disease, and it was under control in the 1960s, with fewer than 5,000 cases reported in the whole continent. The disease resurged at the end of the 1990s, but renewed efforts from endemic countries, cooperation agencies, and nongovernmental organizations led by the World Health

Organization succeeded to raise awareness and resources, while reinforcing national programs, reversing the trend of the cases reported, and bringing the disease under control again. In this context, sustainable elimination of the gambiense HAT, defined as the interruption of the transmission of the disease, was considered as a feasible target for 2030. Since rhodesiense HAT is a zoonosis, where the animal reservoir plays a key role, the interruption of the disease's transmission is not deemed feasible.

Human African trypanosomiasis: the current situation in endemic regions and the risks for non-endemic regions from imported cases.

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Human African trypanosomiasis (HAT) is caused by *Trypanosoma brucei gambiense* and *Trypanosoma brucei rhodesiense* and caused devastating epidemics during the 20th century. Due to effective control programs implemented in the last two decades, the number of reported cases has fallen to a historically low level. Although fewer than 977 cases were reported in 2018 in endemic countries, HAT is still a public health problem in endemic regions until it is completely eliminated. In addition, almost 150 confirmed HAT cases were reported in non-endemic countries in the last three decades. The majority of non-endemic HAT cases were reported in Europe, USA and South Africa, due to historical alliances, economic links or geographic proximity to disease-endemic countries. Furthermore, with the implementation of the 'Belt and Road' project, sporadic imported HAT cases have been reported in China as a warning sign of tropical diseases prevention. In this paper, we explore and interpret the data on HAT incidence and find no positive correlation between the number of HAT cases from endemic and non-endemic countries. This data will provide useful information for better understanding the imported cases of HAT globally in the post-elimination phase.
